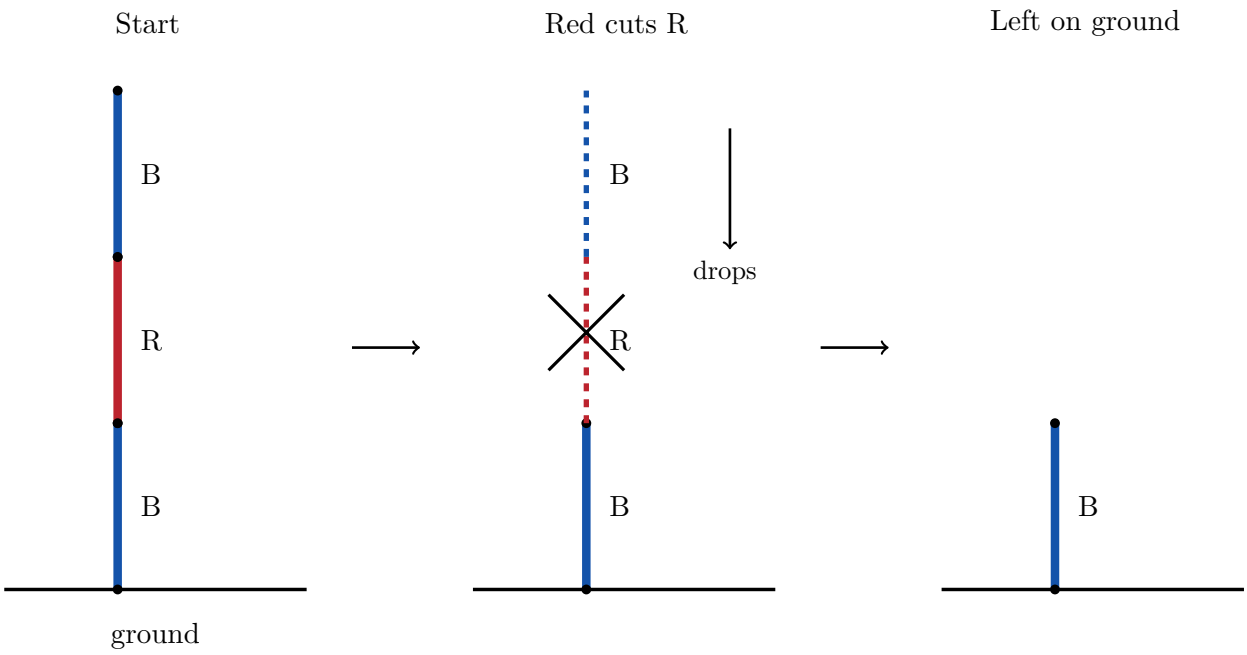
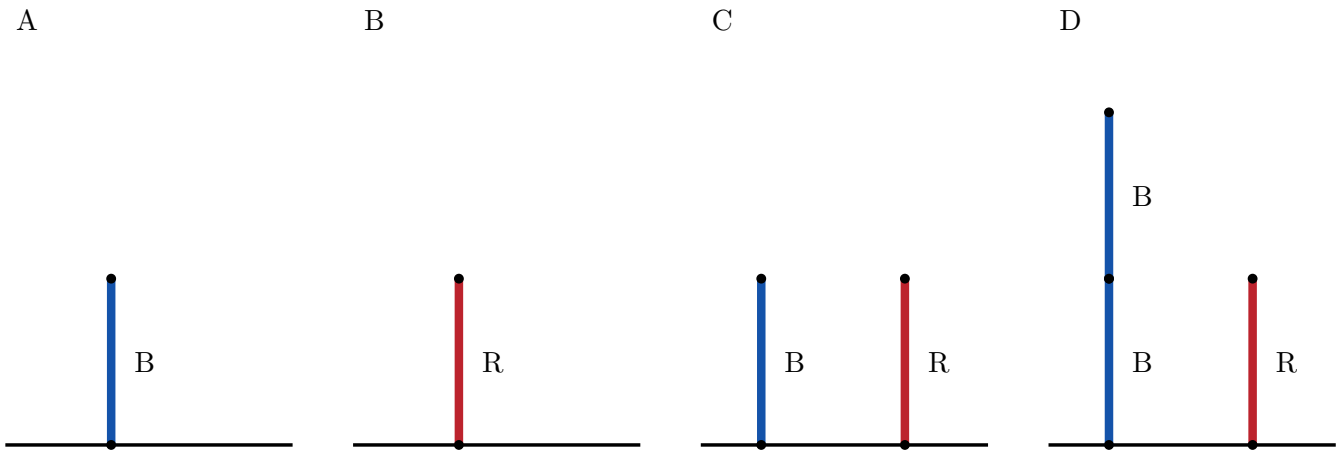
Blue removes one blue (B) segment. Red removes one red (R) segment. All segments cut off from the ground disappear. Take turns; change one stalk per turn. If you cannot move, you lose.

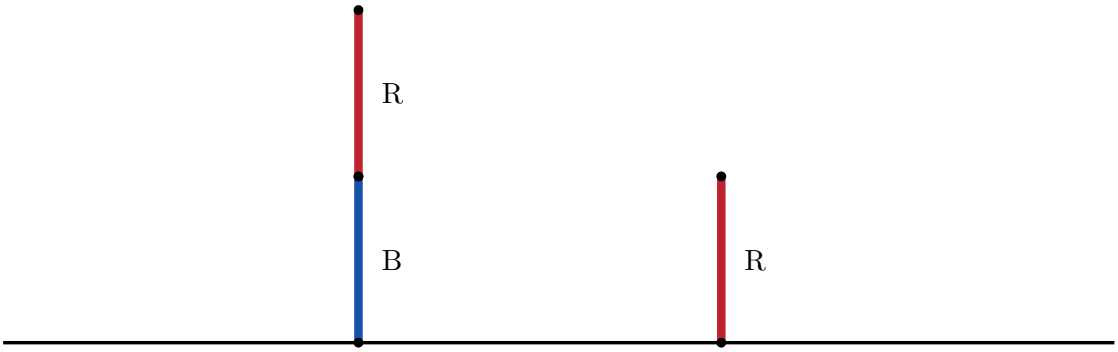


Problem 1: Play each board with Blue starting, then with Red starting. On which boards can one color always win?

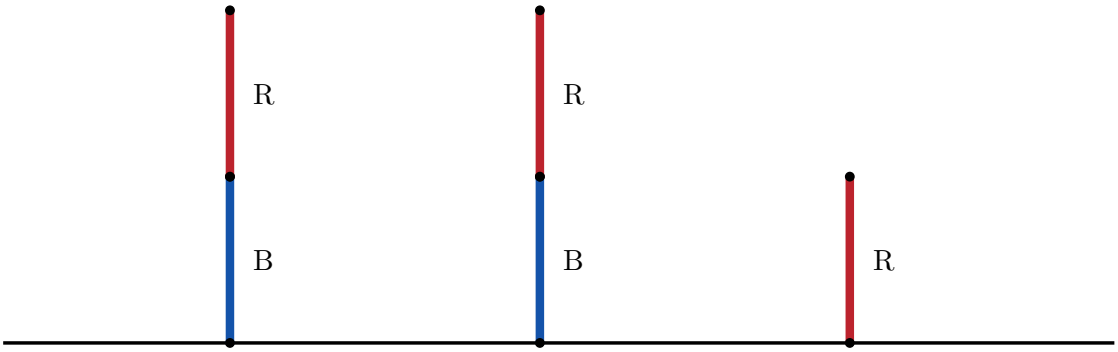


Problem 2: A board balances when the second player can always win, whichever color starts. Which of these boards balance?

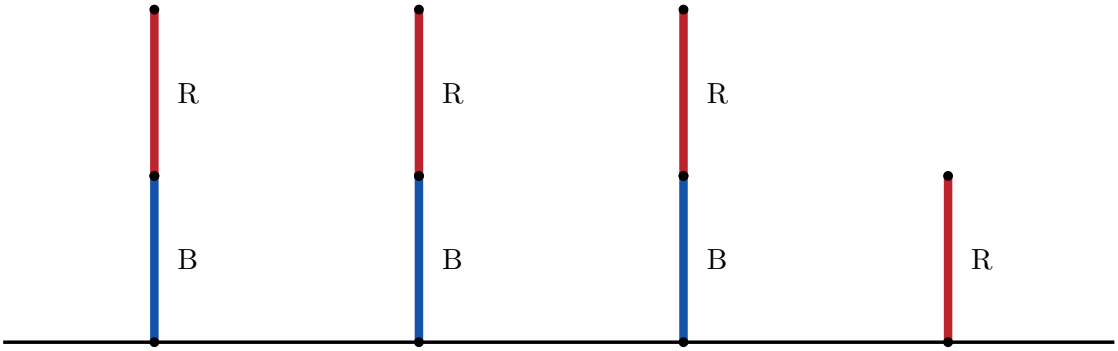
A



B

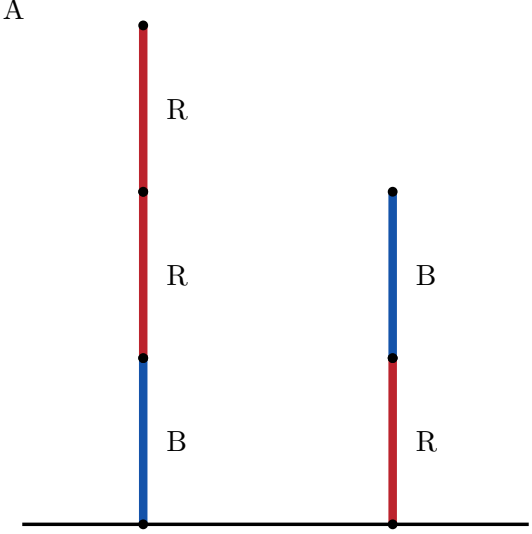


C

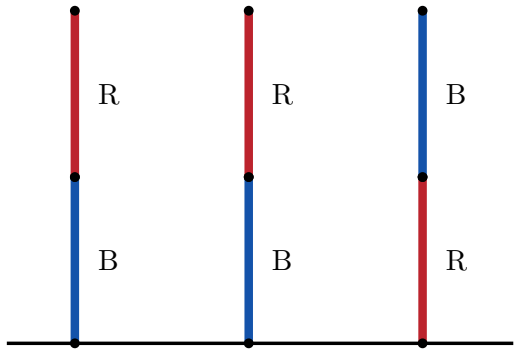


Problem 3: Which of these boards balance? Find a way for the winning color to win every other board, whoever starts.

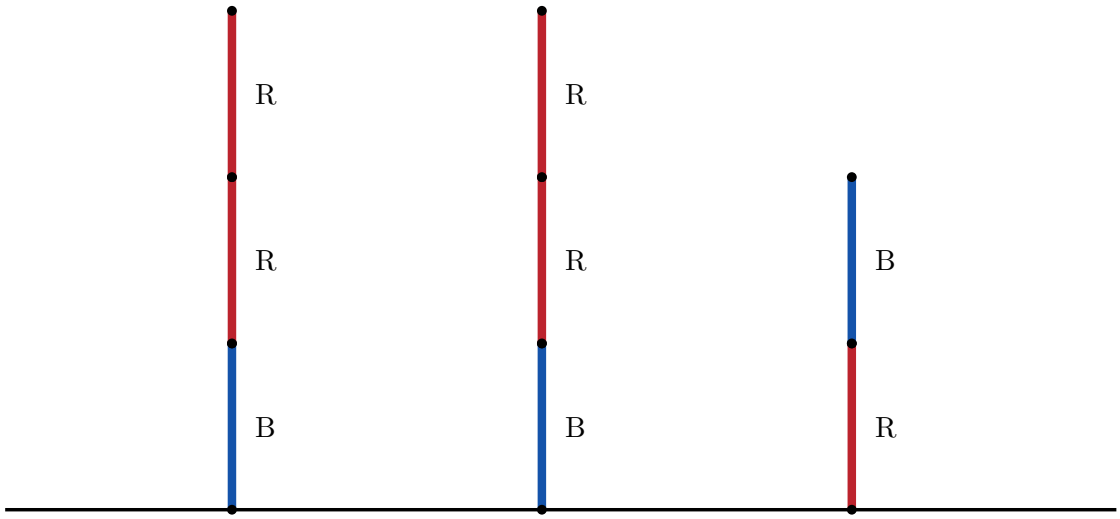
A



B

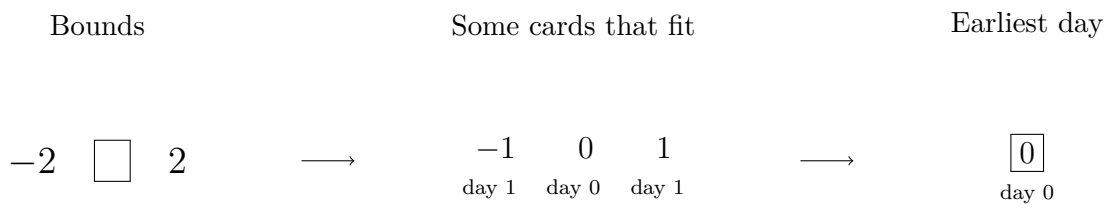


C



Problem 4: Make two different boards that balance and are not in Problems 1–3. Each board must include a stalk with at least two segments.

A number must be greater than every card on its left and less than every card on its right. Choose the card from the earliest day. You may reuse cards.



Problem 5: Find the card each pair of bounds chooses, and put it in the middle.

greater than		less than
0		3

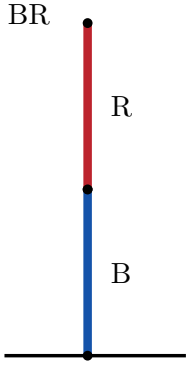
greater than		less than
-1		1

greater than		less than
0		1

greater than		less than
$\frac{1}{2}$		1

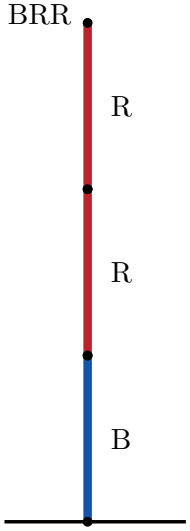
Problem 6: An empty board is 0, B is 1 and R is -1. On each mat, put the numbers of the boards Blue's cuts can leave on the left and Red's on the right. Which card is BR? BRR? RB?

BR



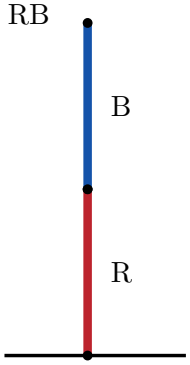
greater than ☐ less than

BRR



greater than ☐ less than

RB



greater than ☐ less than

Number cards

0 day 0	-1 day 1	1 day 1	-2 day 2
$-\frac{1}{2}$ day 2	$\frac{1}{2}$ day 2	2 day 2	-3 day 3
$-\frac{3}{2}$ day 3	$-\frac{3}{4}$ day 3	$-\frac{1}{4}$ day 3	$\frac{1}{4}$ day 3
$\frac{3}{4}$ day 3	$\frac{3}{2}$ day 3	3 day 3	

Extra workspace